5.1 Exponential Functions and Their Graphs

Name _____ Period ____ Date _____

Pre-Calculus 5.1 · Student sheet

I can…

- evaluate and graph exponential functions.
- use the One-to-One Property to solve simple exponential equations.
- use the compound interest formulas.

Vocabulary · fill in the missing word

Exponential function — $f(x) =$ _____, where $a > 0$ and $a \neq 1$

Natural base e — the _____ number $e \approx 2.71828$

Horizontal asymptote — a horizontal line the graph gets _____ and closer to

Part 1 · Grow and shrink

Fill in the table, then graph both functions on one grid.

x	−3	−2	−1	0	1	2	3
2^x							
$(1/2)^x$							

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Use a calculator. Round to four decimal places.

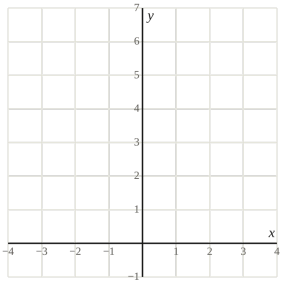
a. $f(x) = 2^x$ at $x = -3.1$

b. $g(x) = e^x$ at $x = -0.32$

Graphs of $f(x) = a^x$: y-intercept (0, 1), horizontal asymptote $y =$ _____; increasing if $a > 1$, decreasing if _____.

Example 2

Graph $f(x) = 4^x$ and $g(x) = 4^{-x}$ on the same grid. Describe each.



One-to-One Property: $a^x = a^y$ if and only if _____.

Example 3

Solve.

a. $9 = 3^{x+1}$

b. $(\frac{1}{2})^x = 8$

Transformations: shifts and reflections work as in P.8; the asymptote moves with _____ shifts.

Example 4

Describe each graph compared with $f(x) = 3^x$. Give the asymptote.

a. $g(x) = 3^{x+1}$

b. $h(x) = 3^x - 2$

c. $k(x) = -3^x$

Compound interest: $A = P(1 + r/n)^{nt}$; compounded continuously, $A =$ _____.

Example 5

$12,000 is invested at 3% annual interest for 5 years. Find the balance if it is compounded (a) monthly and (b) continuously.

Part 3 · Partner practice

Partner 1

Solve $2^{x-3} = 16$.

Partner 2

$5,000 is invested at 4% compounded quarterly for 10 years. Find the balance.